When does a river forecast change a decision?

Flood warning at Gandak Barrage and water allocation on the Potomac

Abstract

We replayed two decisions using river and weather records. On the Gandak, three models signalled the September 2026 rise in barrage inflow six hours ahead, one hour after downstream Khadda had already reached its warning level. We then forecast Khadda's water level directly. Across three test monsoons in 2024–2026, adding upstream Triveni readings gave advance warning for 24 of 26 qualifying crossings of 95 m, compared with 15 using the town gauge alone. For the 14 crossings both caught, the median gain was 4.5 hours. False alert hours increased from 49 to 68. On ten matched next-day forecasts, Bihar's official bulletins had a mean absolute stage error of 9 cm, compared with 11 cm for the upstream model. On the Potomac, we forecast next-day water use and compared it with reconstructed municipal withdrawals on 2,041 days. Mean absolute demand error was 34% lower than using the historical average. Conservative planning cut overcommitment from 36.6% to 1.0% of days while leaving another 98,699 m³/day of demand unmet. Both cases measure what the decision gains and what it costs.